



HIGH PERFORMANCE MEETS THERMAL PRECISION

THERMAL ANALYSIS OF 6U VPX BOARD



Figure: Generic 6U VPX Board

Shape Your Idea To Reality
contact@shiretechnik.com

VPX Technology & Our Value Proposition

VPX is the backbone of next-generation embedded computing, designed for rugged aerospace, defence, and industrial applications. With support for high-speed serial data transfer and modular configurations, VPX boards integrate powerful CPUs, GPUs, and FPGAs into compact 3U and 6U form factors.

“Our VPX Board Thermal Analysis Services ensure that high-performance embedded systems operate at peak efficiency under any harsh conditions. By combining advanced simulation tools with deep expertise in thermal engineering, we deliver solutions that reduce risk, optimize performance, and guarantee compliance with stringent industry standards. With us, thermal management isn’t an afterthought—it’s the foundation of mission success.”

Our Services

01 • Comprehensive Thermal Simulations

Steady-state and transient simulation of 6U VPX design to predict the temperature maps of components, heat-spreader, and wedge-lock.

02 • Cooling Strategy Development

Air, conduction, and liquid cooling solutions tailored to your system.

03 • Design Optimization

Recommendations to improve board layout and airflow.

04 • Altitude & Environmental Derating

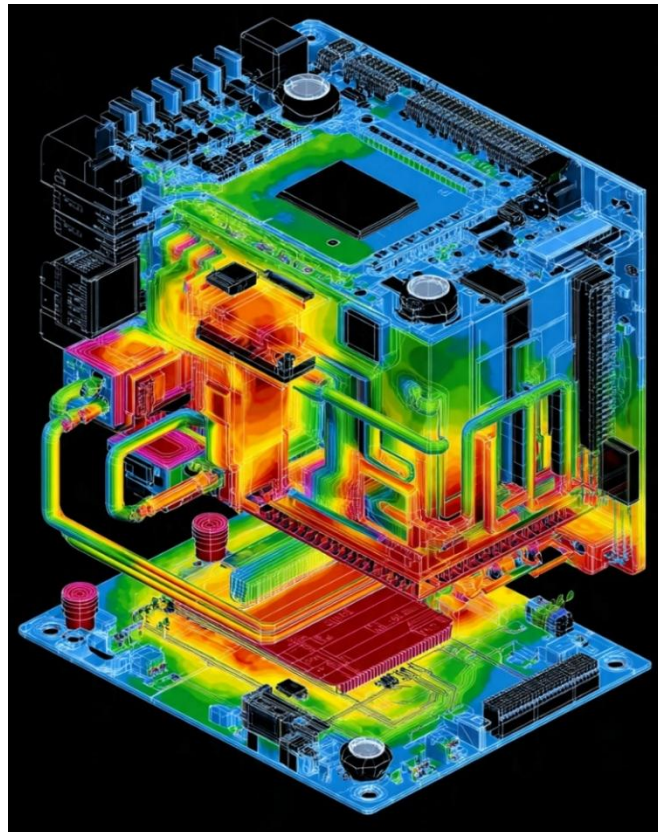
Analysis of reduced convective cooling and pressure effects up to 70,000 ft per the operational envelope and VITA 47 class.

05 • Test Correlation

Chamber and bench test support, including thermal imaging and embedded sensors, to correlate and refine thermal models against measured data.

06 • Compliance Reporting

Documentation aligned with industry standards for certification.



Technical Capabilities

STUDY PARAMETERS	SIMULATION OUTPUT
Component-level insights for CPUs, GPUs, FPGAs, and power regulators	Junction Temperature, Case Temperature
Wedge Lock / Card Edge	Temperature at the VITA 48 wedge lock-to-chassis rail interface
Conduction Rail	Chassis card-cage rail temperature at the card slot
Operating Temperature Range	Functional range across the platform's flight/duty envelope
Altitude Derating	Reduced air density impact on convective cooling (for Air-cooled variants)
Power Dissipation per Slot	Dissipation of total heat load per Slot
Thermal Cycling Endurance	Repeated cycling fatigue on solder joints/BGA interfaces

Compliance & Standard Verification

Verification against VITA & Military cooling standards

- VITA 46 (VPX) — Base mechanical, electrical, and thermal architecture for VPX plug-in modules and backplanes.
- VITA 48 (REDI) — Ruggedized Enhanced Design Implementation; conduction-cooled, air-cooled, and liquid-flow-through card mechanicals.
- VITA 65 (OpenVPX) — System-level architecture defining slot profiles, chassis, and thermal management conventions.
- VITA 47 — Environmental test classes and limits for VPX and VME embedded computing modules.
- SOSA Technical Standard — Sensor Open Systems Architecture alignment for modular, thermally-managed payloads.
- RTCA DO-160 (Section 4 & 5) — Environmental Conditions and Test Procedures for Airborne Equipment: Temperature, Altitude, and Temperature Variation.
- MIL-STD-810H — Environmental Engineering Considerations (Method 501.7 High Temp, 502.7 Low Temp, 503.7 Temp Shock).
- Chassis/Backplane OEM Interface Control Document (ICD) — slot-specific thermal & mechanical requirements.

Deliverables

Thermal Analysis Report (TAR) documenting

Simulation Modelling, Inputs, Assumptions & Boundary conditions

Simulation Results: Heat maps, temperature distribution plots, airflow patterns etc.

Simulation dashboards with performance metrics

Executive summary highlighting risks and recommendations

Design Recommendations

- PCB layout adjustments (component spacing, copper thickness, via placement)
- Heat sink or cold plate design suggestions
- Thermal interface material (TIM) selection guidance etc.

Risk Assessment

- Identification of potential failure points due to overheating
- Impact on system reliability and lifespan
- Mitigation strategies etc.

Compliance-ready documentation for certification

Engagement Model

- **Requirement Analysis:** Scoping workshop to capture the card's power dissipation map, duty cycle, slot position, chassis/backplane installation details etc.
- **Pre-Simulation Review:** CFD/FE model build, heat path, boundary condition definition, and analysis execution.
- **Pre-Report:** Draft TAR issued for customer review, followed by chamber/bench test correlation (if required).
- Final certification-ready TAR delivered with margin summary and compliance mapping.



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